

## Chapter 1: Natural Resources – Air, Water and Land

Textbook Exercise Solutions & Answers

### 1. Fill in the blanks and rewrite the completed statements.

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(a) The layer of ozone gas absorbs ..... rays that come from the sun to the earth.

Ultraviolet (UV) rays

(b) Of the total water available on the earth, fresh water forms ..... percent.

0.3 %

(c) Both ..... and ..... constituents are present in the soil.

biotic, abiotic

### 2. why is it said that?

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(a) The ozone layer is a protective shell of earth.

1. The ultra violet (UV) rays coming from the sun are very harmful for living things.
2. The ozone layer present in the lower stratosphere absorb this UV. rays and prevent them from reaching the earth. As a result life on earth is protected.
3. Therefore, it is said that the ozone layer is a protective shell of the earth.

(b) Water is life.

1. Water is a good solvent and it dissolves many substances.
2. The human blood is made of 70% water and the sap of plant also contains a very high proportion of water.
3. All the life processes would not take place in the absence of water.
4. Hence, without water no living organism can survive.
5. Therefore, it is said that 'water is life'.

(c) Sea water is useful even though it is not potable. OR In what way is sea water useful even though it is salty?

1. Many fish and aquatic animals live in sea water.
2. The water from the sea evaporates to form clouds which brings rain.
3. The sea water also helps the land to cool due to breezes.
4. The salt and minerals are also obtained from sea water.
5. Corals and pearls are obtained from sea animals.
6. Thus, sea water is useful even though it is not potable.

### 3. What will happen if

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(a)

(b) Microbes in soil get destroyed.

1. Microbes in the soil decompose dead plants and animals and convert it into humus. This humus supplies nutrients to the soil.
2. Humus also aerates soil and holds water in it. It makes the soil more fertile.
3. If microbes are destroyed, humus will not be formed and the soil will not become fertile, making it unsuitable for growth of plants. Also dead and decaying matter will accumulate on land.

**(c) The number of vehicles and factories in your surroundings increases.**

1. Vehicles and factories are the major cause of air pollution.
2. They release harmful gases like carbon dioxide, carbon monoxide and sulphur dioxide into the air.
3. These pollutants are harmful to the environment and to the people living in the surrounding area.
4. Hence, if the number of vehicles and factories in our surroundings increases, the air pollution will also increase.

**(d) The total supply of potable water is finished.**

1. Water plays very important role in the survival of living organism.
2. All living things are dependent on water.
3. A very small quantity of water is potable and can be used for drinking.
4. All bodily functions are regulated by water.
5. Therefore, if total supply of potable water is finished, plants and animals will not survive and there will be no life on earth.

**4. Match the following.**

Group 'A' Group 'B' 1. Carbon dioxide a. Generation of soil 2. Oxygen b. Rain 3. Water vapour c. Plants and food production 4. Microbes d. Combustion

| Group 'A'         | Group 'B'                     |
|-------------------|-------------------------------|
| 1. Carbon dioxide | c. Plants and food production |
| 2. Oxygen         | d. Combustion                 |
| 3. Water vapour   | b. Rain                       |
| 4. Microbes       | a. Generation of soil         |

**5. Name the following.**

**(a) Constituents of biosphere.**

Atmosphere, hydrosphere, lithosphere and all living things on earth.

**(b) Biotic constituents of soil.**

Microbes, worms, insects, burrowing rodents like rats, mice, roots of trees and plants.

**(c) Fossil fuel.**

Crude oil from which we get kerosene, petrol, diesel, paraffin wax and tar.

**(d) Inert gases in air.**

Neon, argon, helium, krypton, xenon.

**(e) Gases that are harmful to ozone layer.**

Chlorofluorocarbon and carbon tetrachloride.

**6. True or False?**

**(a) Land and soil is the same thing.**

False – Land consists of stones, soil and big rocks.

**(b) The water in a lake is called ground water.**

False – Water trapped below the ground over the bedrocks is called ground water.

(c) It takes about thousand years to form a 25 cm thick layer of soil.

False – It almost takes around thousand years to form a 2.5 cm thick layer of soil.

(d) Radon is used in decorative lights.

False – Neon is used in decorative lights.

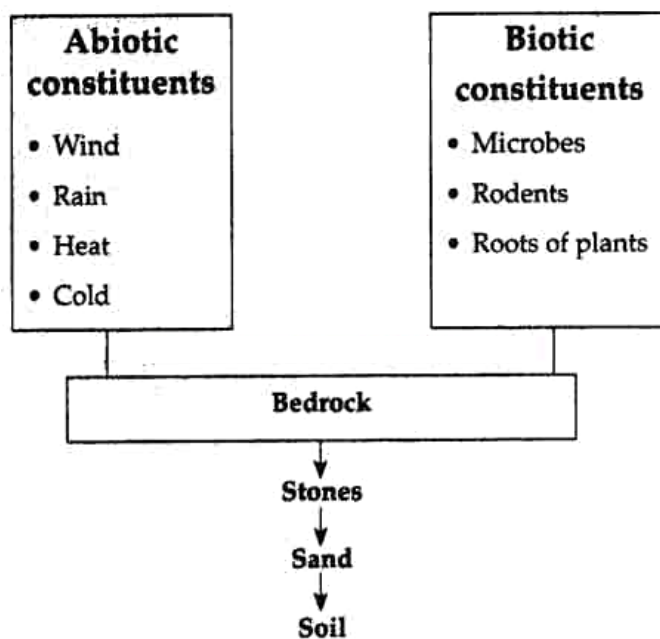
## 7. Answer in your own words.

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(a) Explain with the help of a diagram how soil is formed.

1. The soil on the land is formed by a natural process.
2. The abiotic components of soil are supplied through the weathering of the bedrock.
3. Due to heat, cold wind and rain the bedrock breaks down into pieces.
4. Stones, sand and soil are formed from these pieces.
5. Microbes, worms, insects, rodents and roots of trees growing on land help in weathering of rocks.

### Formation of Soil



This process is slow, continuous and it takes a thousand years to form 2.5 cm thick layer of mature soil.

(b) Why is there a shortage of water even though it occupies about 71% of the earth's surface?

1. 71% of earth's surface is covered with water of, which 97% is salty water present in seas and oceans and 2.7% water is available as ground water, ice and in other forms.
2. Only 0.3% water is available as fresh water which can be used for drinking.
3. All the living organisms require water to drink. The water is used in the industry and also for farming.
4. Due to increasing population and uncontrolled usage, we experience shortage of water.

(c) What are the various constituents of air? Write their uses.

Air contains gases like oxygen, carbon dioxide, nitrogen, inert gases, water vapour and dust particles. The uses of constituents of air are as follows.

|     | Gases                                                                          | Proportion in atmosphere | Uses                                                                                                                   |
|-----|--------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------|
| (a) | Nitrogen                                                                       | 78%                      | (i) Helps in building necessary proteins.<br>(ii) In production of ammonia.<br>(iii) Air tight packaging of foodstuff. |
| (b) | Oxygen                                                                         | 21%                      | (i) Used in combustion.<br>(ii) Necessary for respiration.                                                             |
| (c) | Carbon Dioxide                                                                 | 0.03%                    | (i) Used in fire extinguishers.<br>(ii) Used in plant food preparation.                                                |
| (d) | Inert Gases<br>(1) Argon<br>(2) Helium<br>(3) Neon<br>(4) Krypton<br>(5) Xenon | 0.9%                     | (i) Electric bulb<br>(ii) Used in airship<br>(iii) Street light<br>(iv) Fluorescence tubes<br>(v) Flash photography    |
| (e) | Other constituents like dust, water vapour                                     | 0.7%                     | (i) Used in formation of cloud, fog, dew etc.<br>(ii) Spread sunlight in all directions.                               |

(d) Why are air, water and land considered to be valuable natural resources?

1. The various components of air help and support the growth of living organisms.
2. Nitrogen is used to make proteins, oxygen is used for respiration and carbon dioxide is used to make food by plants.
3. Similarly, land supports growth of terrestrial plants and animals. It provides important minerals to plants and also to human beings.
4. It is the shelter for worms, insects and rodents and supports their growth. Plants also cannot survive without land.
5. Water is necessary for carrying out all life processes in the living organisms and without water there will be no life on earth.
6. Hence land, air and water are considered valuable natural resources.